

Linear Regression Lecture

Make sure you've watched the intro video on Linear Regression and read Chapters 2 and 3 of ISLR before continuing!

Remember that Linear Regression is a supervised learning algorithm, meaning we'll have labeled data and try to predict new labels on unlabeled data. We'll explore some of the following concepts:

- Get our Data
- Exploratory Data Analysis (EDA)
- Clean our Data
- Review of Model Form
- Train and Test Groups
- Linear Regression Model

Get our Data

We will use the [Student Performance Data Set from UC Irvine's Machine Learning Repository!](#) Download this data or just use the supplied csv files in the notebook repository. We'll specifically look at the math class (student-mat.csv). Make sure to take note that the delimiter is a semi-colon.

In [1]:

```
# Read CSV, note the delimiter (sep)
df <- read.csv('student-mat.csv', sep=';')
```

In [2]:

```
head(df)
```

Out[2]:

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	reason	guardian	travelttime	studyin
1	GP	F	18	U	GT3	A	4	4	at_home	teacher	course	mother	2	2
2	GP	F	17	U	GT3	T	1	1	at_home	other	course	father	1	2
3	GP	F	15	U	LE3	T	1	1	at_home	other	other	mother	1	2
4	GP	F	15	U	GT3	T	4	2	health	services	home	mother	1	3
5	GP	F	16	U	GT3	T	3	3	other	other	home	father	1	2
6	GP	M	16	U	LE3	T	4	3	services	other	reputation	mother	1	2

In [3]:

```
summary(df)
```

Out[3]:

school	sex	age	address	famsize	Pstatus	Medu
GP:349	F:208	Min.:15.0	R: 88	GT3:281	A: 41	Min.:0.000
MS: 46	M:187	1st Qu.:16.0	U:307	LE3:114	T:354	1st Qu.:2.000
		Median:17.0				Median:3.000
		Mean:16.7				Mean:2.749
		3rd Qu.:18.0				3rd Qu.:4.000
		Max.:22.0				Max.:4.000
Fedu	Mjob	Fjob	reason	guardian		
Min.:0.000	at_home:59	at_home:20	course:145	father:90		
1st Qu.:2.000	health:34	health:18	home:109	mother:273		
Median:2.000	other:141	other:217	other:36	other:32		
Mean:2.522	services:103	services:111	reputation:105			
3rd Qu.:3.000	teacher:58	teacher:29				
Max.:4.000						

traveltime	studytime	failures	schoolsup	famsup	paid
Min. :1.000	Min. :1.000	Min. :0.0000	no :344	no :153	no :214
1st Qu.:1.000	1st Qu.:1.000	1st Qu.:0.0000	yes: 51	yes:242	yes:181
Median :1.000	Median :2.000	Median :0.0000			
Mean :1.448	Mean :2.035	Mean :0.3342			
3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:0.0000			
Max. :4.000	Max. :4.000	Max. :3.0000			

activities	nursery	higher	internet	romantic	famrel
no :194	no : 81	no : 20	no : 66	no :263	Min. :1.000
yes:201	yes:314	yes:375	yes:329	yes:132	1st Qu.:4.000
					Median :4.000
					Mean :3.944
					3rd Qu.:5.000
					Max. :5.000

freetime	goout	Dalc	Walc
Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.:3.000	1st Qu.:2.000	1st Qu.:1.000	1st Qu.:1.000
Median :3.000	Median :3.000	Median :1.000	Median :2.000
Mean :3.235	Mean :3.109	Mean :1.481	Mean :2.291
3rd Qu.:4.000	3rd Qu.:4.000	3rd Qu.:2.000	3rd Qu.:3.000
Max. :5.000	Max. :5.000	Max. :5.000	Max. :5.000

health	absences	G1	G2
Min. :1.000	Min. : 0.000	Min. : 3.00	Min. : 0.00
1st Qu.:3.000	1st Qu.: 0.000	1st Qu.: 8.00	1st Qu.: 9.00
Median :4.000	Median : 4.000	Median :11.00	Median :11.00
Mean :3.554	Mean : 5.709	Mean :10.91	Mean :10.71
3rd Qu.:5.000	3rd Qu.: 8.000	3rd Qu.:13.00	3rd Qu.:13.00
Max. :5.000	Max. :75.000	Max. :19.00	Max. :19.00

G3
Min. : 0.00
1st Qu.: 8.00
Median :11.00
Mean :10.42
3rd Qu.:14.00
Max. :20.00

Attribute Information

Here is the attribute information for our data set: Attribute Information:

Attributes for both student-mat.csv (Math course) and student-por.csv (Portuguese language course) datasets:

- 1 school - student's school (binary: 'GP' - Gabriel Pereira or 'MS' - Mousinho da Silveira)
- 2 sex - student's sex (binary: 'F' - female or 'M' - male)
- 3 age - student's age (numeric: from 15 to 22)
- 4 address - student's home address type (binary: 'U' - urban or 'R' - rural)
- 5 famsize - family size (binary: 'LE3' - less or equal to 3 or 'GT3' - greater than 3)
- 6 Pstatus - parent's cohabitation status (binary: 'T' - living together or 'A' - apart)
- 7 Medu - mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)
- 8 Fedu - father's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)
- 9 Mjob - mother's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')
- 10 Fjob - father's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')
- 11 reason - reason to choose this school (nominal: close to 'home', school 'reputation', 'course' preference or 'other')
- 12 guardian - student's guardian (nominal: 'mother', 'father' or 'other')
- 13 traveltime - home to school travel time (numeric: 1 - less than 15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - more than 1 hour)
- 14 studytime - weekly study time (numeric: 1 - less than 2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - more than 10 hours)
- 15 failures - number of past class failures (numeric: n if between 1 and 3 , else 4)
- 16 schoolsup - extra educational support (binary: yes or no)
- 17 famsup - family educational support (binary: yes or no)
- 18 paid - extra paid classes within the course subject (Math or Portuguese) (binary: yes or no)
- 19 activities - extra-curricular activities (binary: yes or no)
- 20 nursery - attended nursery school (binary: yes or no)
- 21 higher - wants to take higher education (binary: yes or no)
- 22 internet - Internet access at home (binary: yes or no)

- 22 internet - internet access at home (binary: yes or no)
- 23 romantic - with a romantic relationship (binary: yes or no)
- 24 famrel - quality of family relationships (numeric: from 1 - very bad to 5 - excellent)
- 25 freetime - free time after school (numeric: from 1 - very low to 5 - very high)
- 26 goout - going out with friends (numeric: from 1 - very low to 5 - very high)
- 27 Dalc - workday alcohol consumption (numeric: from 1 - very low to 5 - very high)
- 28 Walc - weekend alcohol consumption (numeric: from 1 - very low to 5 - very high)
- 29 health - current health status (numeric: from 1 - very bad to 5 - very good)
- 30 absences - number of school absences (numeric: from 0 to 93)

these grades are related with the course subject, Math or Portuguese:

- 31 G1 - first period grade (numeric: from 0 to 20)
- 31 G2 - second period grade (numeric: from 0 to 20)
- 32 G3 - final grade (numeric: from 0 to 20, output target)

Clean the Data

Next we have to clean this data. This data is actually already cleaned for you, But here are some things you may want to consider doing for other data sets:

Check for NA values

Let's see if we have any NA values:

In [4]:

```
any(is.na(df))
```

Out[4]:

FALSE

Great! Most real data sets will probably have NA or Null values, so its always good to check! Its up to you how to deal with them, either dropping them if they aren't too many, or imputing other values, like the mean value.

Categorical Features

Moving on, let's make sure that categorical variables have a factor set to them. For example, the MJob column refers to categories of Job Types, not some numeric value from 1 to 5. R is actually really good at detecting these sort of values and will take of this work for you a lot of the time, but always keep in mind the use of **factor()** as a possible. Luckily this is basically already, we can check this using the **str()** function:

In [5]:

```
str(df)
```

```
'data.frame': 395 obs. of 33 variables:
 $ school : Factor w/ 2 levels "GP","MS": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex : Factor w/ 2 levels "F","M": 1 1 1 1 1 2 2 1 2 2 ...
 $ age : int 18 17 15 15 16 16 16 17 15 15 ...
 $ address : Factor w/ 2 levels "R","U": 2 2 2 2 2 2 2 2 2 2 ...
 $ famsize : Factor w/ 2 levels "GT3","LE3": 1 1 2 1 1 2 2 1 2 1 ...
 $ Pstatus : Factor w/ 2 levels "A","T": 1 2 2 2 2 2 2 1 1 2 ...
 $ Medu : int 4 1 1 4 3 4 2 4 3 3 ...
 $ Fedu : int 4 1 1 2 3 3 2 4 2 4 ...
 $ Mjob : Factor w/ 5 levels "at_home","health",...: 1 1 1 2 3 4 3 3 4 3 ...
 $ Fjob : Factor w/ 5 levels "at_home","health",...: 5 3 3 4 3 3 3 5 3 3 ...
 $ reason : Factor w/ 4 levels "course","home",...: 1 1 3 2 2 4 2 2 2 2 ...
 $ guardian : Factor w/ 3 levels "father","mother",...: 2 1 2 2 1 2 2 2 2 2 ...
 $ traveltime: int 2 1 1 1 1 1 1 2 1 1 ...
 $ studytime : int 2 2 2 3 2 2 2 2 2 2 ...
 $ failures : int 0 0 3 0 0 0 0 0 0 0 ...
 $ schoolsup : Factor w/ 2 levels "no","yes": 2 1 2 1 1 1 1 2 1 1 ...
 $ famsup : Factor w/ 2 levels "no","yes": 1 2 1 2 2 2 1 2 2 2 ...
 $ paid : Factor w/ 2 levels "no","yes": 1 1 2 2 2 2 1 1 2 2 ...
 $ activities: Factor w/ 2 levels "no","yes": 1 1 1 2 1 2 1 1 1 2 ...
```

```

$ nursery      : Factor w/ 2 levels "no","yes": 2 1 2 2 2 2 2 2 2 ...
$ higher       : Factor w/ 2 levels "no","yes": 2 2 2 2 2 2 2 2 2 ...
$ internet     : Factor w/ 2 levels "no","yes": 1 2 2 2 1 2 2 1 2 2 ...
$ romantic     : Factor w/ 2 levels "no","yes": 1 1 1 2 1 1 1 1 1 1 ...
$ famrel       : int    4 5 4 3 4 5 4 4 4 5 ...
$ freetime     : int    3 3 3 2 3 4 4 1 2 5 ...
$ goout        : int    4 3 2 2 2 2 4 4 2 1 ...
$ Dalc         : int    1 1 2 1 1 1 1 1 1 1 ...
$ Walc         : int    1 1 3 1 2 2 1 1 1 1 ...
$ health       : int    3 3 3 5 5 5 3 1 1 5 ...
$ absences     : int    6 4 10 2 4 10 0 6 0 0 ...
$ G1           : int    5 5 7 15 6 15 12 6 16 14 ...
$ G2           : int    6 5 8 14 10 15 12 5 18 15 ...
$ G3           : int    6 6 10 15 10 15 11 6 19 15 ...

```

Exploratory Data Analysis

Let's use ggplot2 to explore the data a bit. Feel free to expand on this section:

In [6]:

```

library(ggplot2)
library(ggthemes)
library(dplyr)

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

Correlation and CorrPlots

From Wikipedia, correlation is defined as:

- In statistics, dependence or association is any statistical relationship, whether causal or not, between two random variables or two sets of data. Correlation is any of a broad class of statistical relationships involving dependence, though in common usage it most often refers to the extent to which two variables have a linear relationship with each other. Familiar examples of dependent phenomena include the correlation between the physical statures of parents and their offspring, and the correlation between the demand for a product and its price.

Correlation plots are a great way of exploring data and seeing if there are any interaction terms. Let's start off by just grabbing the numeric data (we can't see correlation for categorical data):

In [7]:

```

# Grab only numeric columns
num.cols <- sapply(df, is.numeric)

# Filter to numeric columns for correlation
cor.data <- cor(df[,num.cols])

cor.data

```

Out[7]:

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	G1
age	1.000000000	-0.163658419	0.163438069	0.070640721	-0.004140037	0.243665377	0.053940096	0.016434389	0.000000000
Medu	-0.163658419	1.000000000	0.623455112	-0.171639305	0.064944137	-0.236679963	-0.003914458	0.030890867	0.000000000

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	
Fedu	0.163438069	0.623455112	1.000000000	0.158194054	0.009174639	0.250408444	0.001369727	0.012845528	
traveltime	0.070640721	-	-	1.000000000	-	0.092238746	-	-	
studytime	-	0.064944137	-	-	1.000000000	-	0.039730704	-	
failures	0.24366538	-0.23667996	-0.25040844	0.09223875	-0.17356303	1.00000000	-0.04433663	0.09198747	
famrel	0.053940096	-	-	-	0.039730704	-	1.00000000	0.150701444	
freetime	0.01643439	0.03089087	-0.01284553	-0.01702494	-0.14319841	0.09198747	0.15070144	1.00000000	
goout	0.126963880	0.064094438	0.043104668	0.028539674	-	0.124560922	0.064568411	0.285018715	
Dalc	0.131124605	0.019834099	0.002386429	0.138325309	-	0.136046931	-	0.209000848	
Walc	0.11727605	-0.04712346	-0.01263102	0.13411575	-0.25378473	0.14196203	-0.11339731	0.14782181	
health	-	-	0.014741537	0.007500606	-	0.065827282	0.094055728	0.075733357	
absences	0.17523008	0.10028482	0.02447289	-0.01294378	-0.06270018	0.06372583	-0.04435409	-0.05807792	
G1	-0.06408150	0.20534100	0.19026994	-0.09303999	0.16061192	-0.35471761	0.02216832	0.01261293	
G2	-0.14347405	0.21552717	0.16489339	-0.15319796	0.13588000	-0.35589563	-0.01828135	-0.01377714	
G3	-0.16157944	0.21714750	0.15245694	-0.11714205	0.09781969	-0.36041494	0.05136343	0.01130724	

While this is fantastic information, it's hard to take it all in. Let's visualize all this data. There are lots of amazing 3rd party packages to do this, let's use and install the 'corrgram' package and the corrplot package. This will also install a bunch of dependencies for the package.

In [8]:

```
#install.packages('corrgram',repos = 'http://cran.us.r-project.org')
#install.packages('corrplot',repos = 'http://cran.us.r-project.org')
```

In [9]:

```
library(corrplot)
library(corrgram)
```

Warning message:
: replacing previous import by 'magrittr::%>%' when loading 'dendextend'

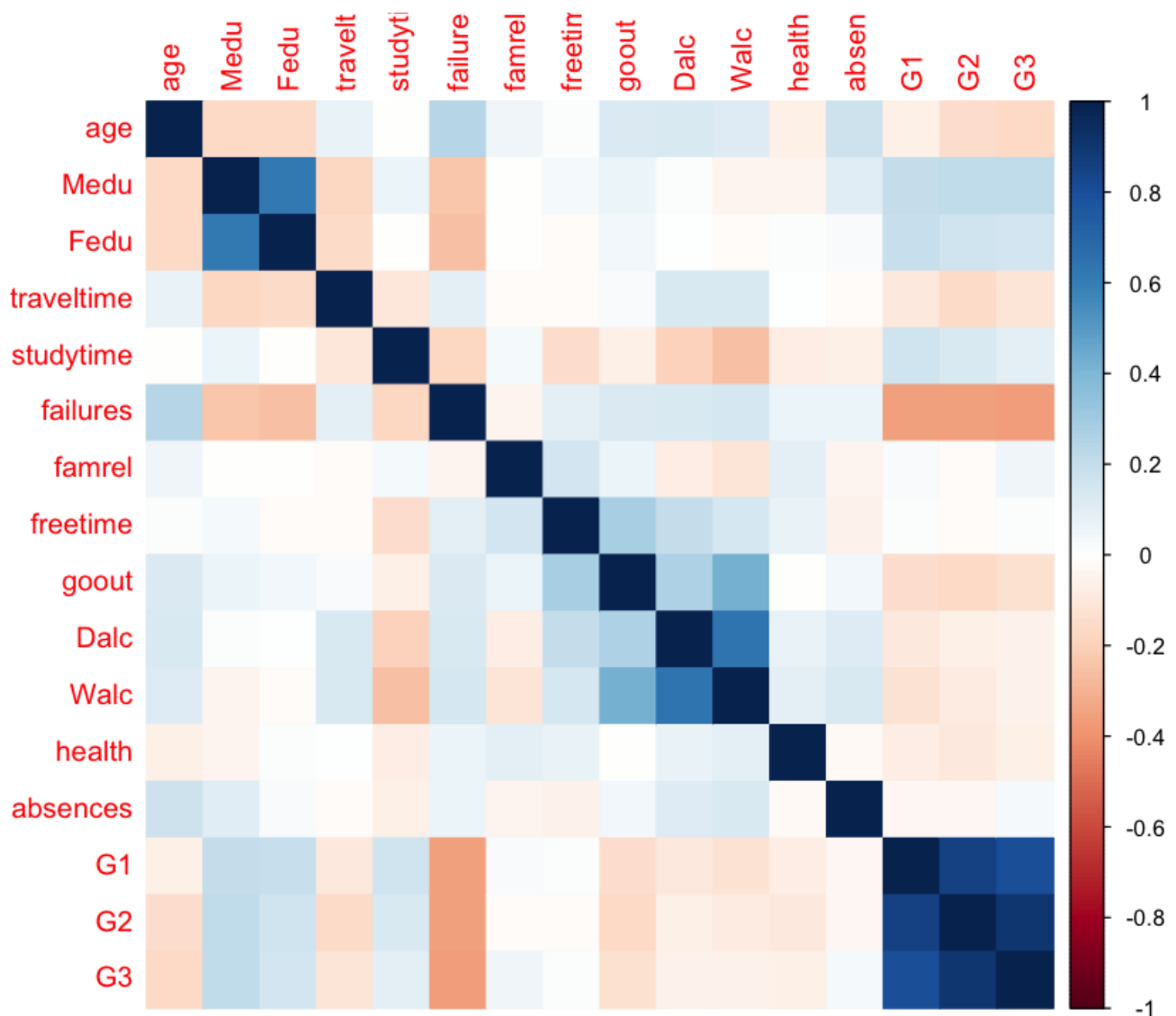
Let's start by using corrplot, the most common one. [Here's a really nice documentation page on the package.](#) I encourage you to play around with it.

In [10]:

```
#help(corrplot)
```

In [11]:

```
corrplot(cor.data,method='color')
```



Clearly we have very high correlation between G1, G2, and G3 which makes sense since those are grades:

- 31 G1 - first period grade (numeric: from 0 to 20)
- 31 G2 - second period grade (numeric: from 0 to 20)
- 32 G3 - final grade (numeric: from 0 to 20, output target)

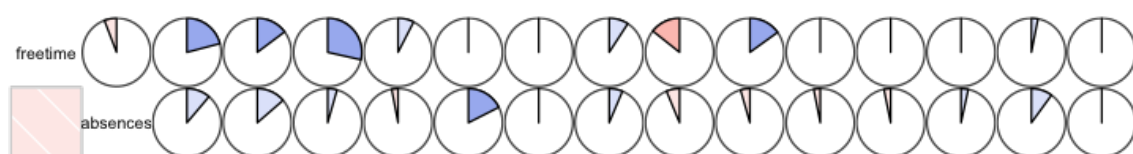
Meaning good students do well each period, and poor students do poorly each period, etc. Also a high G1, G2, or G3 value has a negative correlation with failure (number of past class failures).

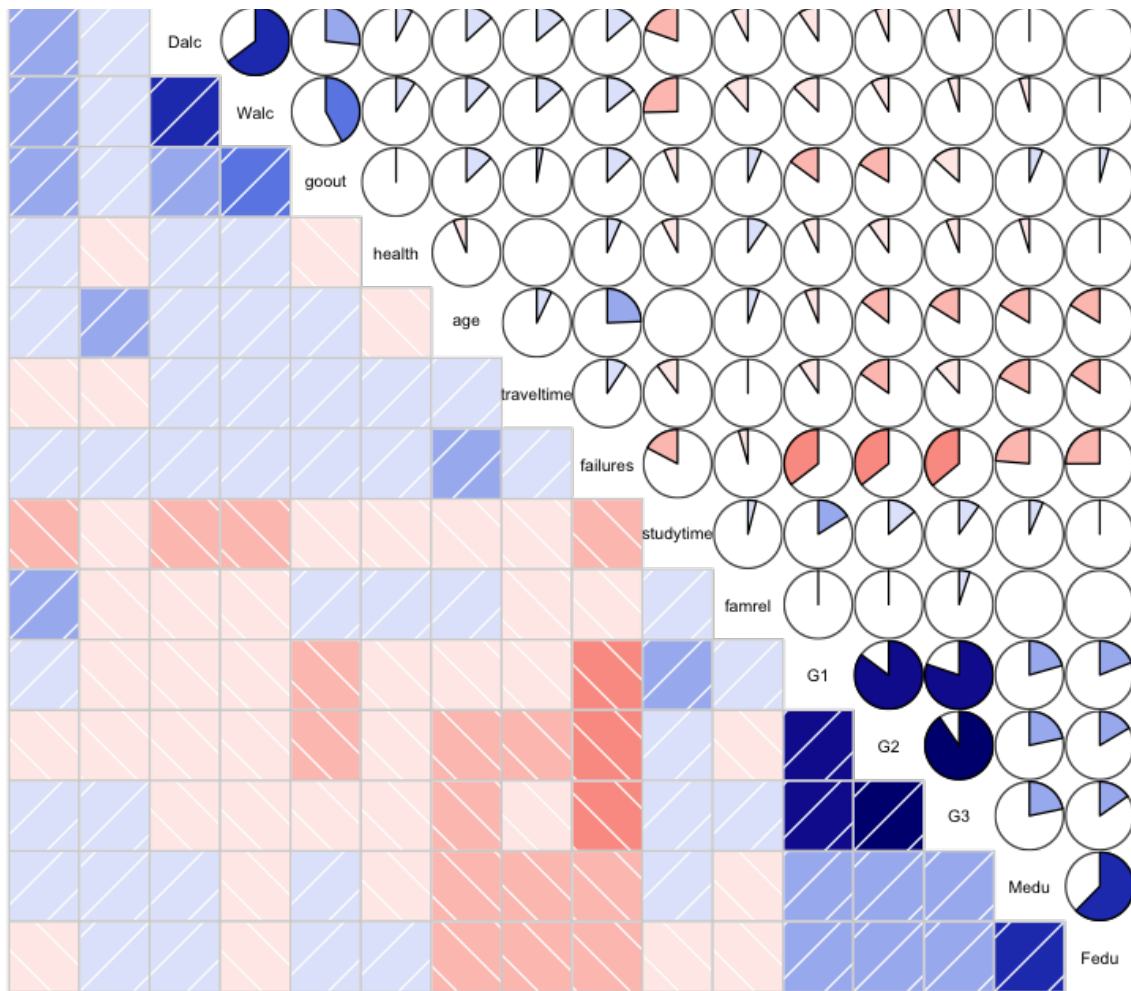
Also Mother and Father education levels are correlated, which also makes sense.

We can also use the `corrgram` which allows to just automatically do these type of figures by just passing in the dataframe directly. There's a lot of going on here, so reference the [documentation of corrgram](#) for more info.

In [12]:

```
corrgram(df, order=TRUE, lower.panel=panel.shade,
         upper.panel=panel.pie, text.panel=panel.txt)
```

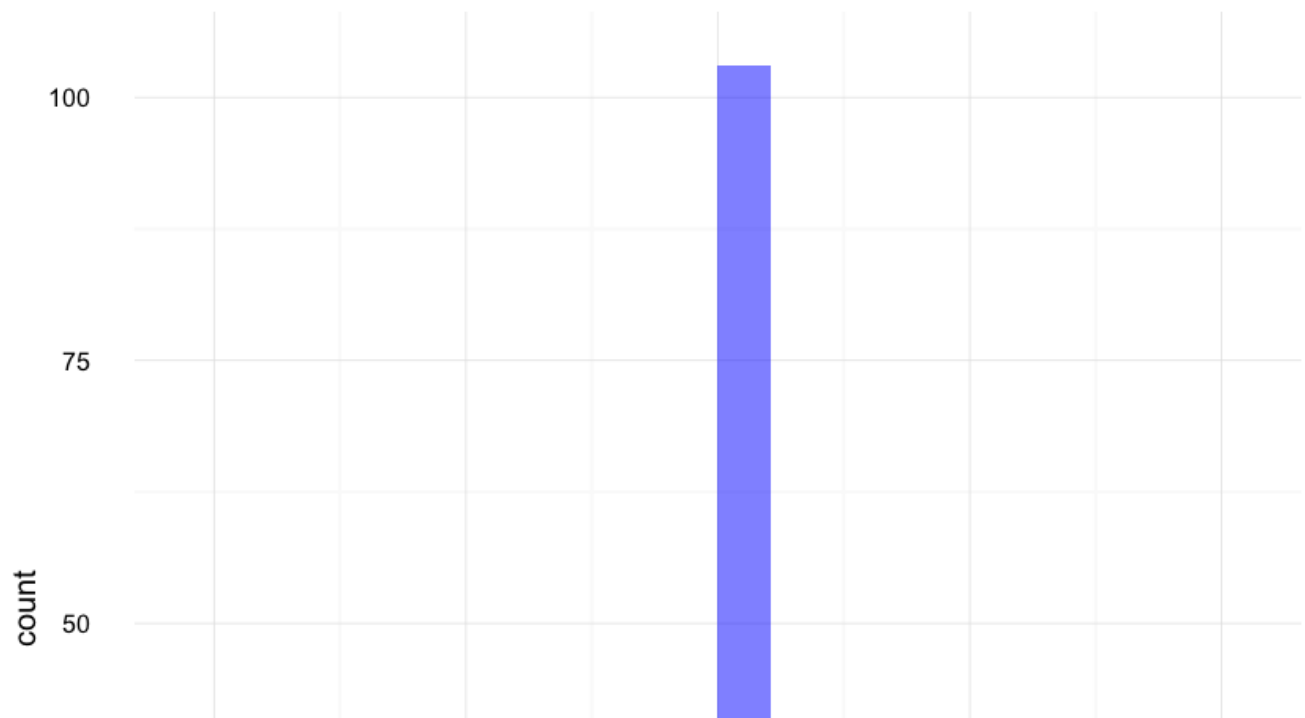


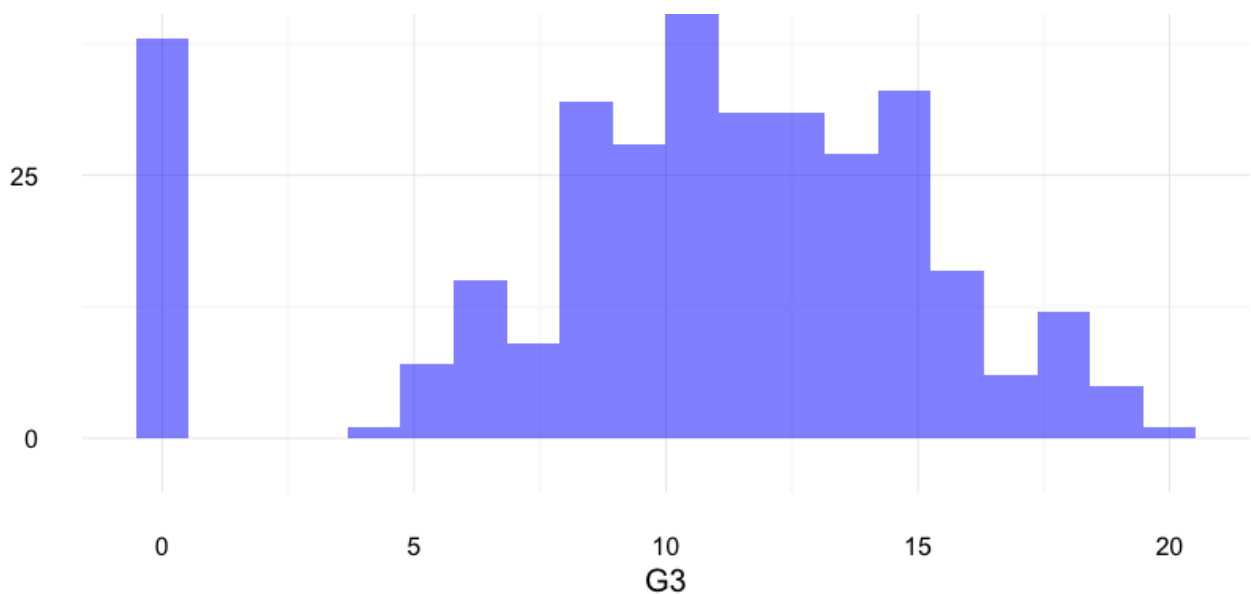


Since we're going to eventually try to predict the G3 score let's see a histogram of these scores:

In [13]:

```
ggplot(df, aes(x=G3)) + geom_histogram(bins=20, alpha=0.5, fill='blue') + theme_minimal()
```





Looks like quite a few students get a zero. This is a good place to ask questions, like are students missing the test? Also why is the mean occurrence so high? Is this test curved?

Let's continue by building a model

Building a Model

General Form:

The general model of building a linear regression model in R looks like this:

```
model <- lm(y ~ x1 + x2,data)
```

or to use all the features in your data

```
model <- lm(y ~. , data) # Uses all features
```

Train and Test Data

We'll need to split our data into a training set and a testing set in order to test our accuracy. We can do this easily using the caTools library:

In [14]:

```
# Import Library
library(caTools)
# Set a random seed so your "random" results are the same as this notebook
set.seed(101)

# Split up the sample, basically randomly assigns a boolean to a new column "sample"
sample <- sample.split(df$age, SplitRatio = 0.70) # SplitRatio = percent of sample==TRUE

# Training Data
train = subset(df, sample == TRUE)

# Testing Data
test = subset(df, sample == FALSE)
```

Training our Model

Let's train our model on our training data, then ask for a summary of that model:

In [15]:


```
In [15]:
```

```
model <- lm(G3 ~ .,train)
```

```
In [16]:
```

```
summary(model)
```

```
Out[16]:
```

```
Call:
```

```
lm(formula = G3 ~ ., data = train)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-7.7681 -0.6423  0.2294  1.0691  4.5942
```

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.329568	2.474569	-0.537	0.591574
schoolMS	0.838581	0.470545	1.782	0.076016 .
sexM	0.034883	0.275586	0.127	0.899382
age	-0.214994	0.119579	-1.798	0.073472 .
addressU	0.067190	0.326035	0.206	0.836905
famsizeLE3	-0.111068	0.283228	-0.392	0.695302
PstatusT	-0.153653	0.401679	-0.383	0.702417
Medu	0.279949	0.171111	1.636	0.103164
Fedu	-0.221275	0.151103	-1.464	0.144422
Mjobhealth	0.002065	0.610532	0.003	0.997304
Mjobother	0.509947	0.403195	1.265	0.207209
Mjobservices	0.475476	0.435332	1.092	0.275857
Mjobteacher	0.285345	0.550640	0.518	0.604802
Fjobhealth	0.433172	0.774191	0.560	0.576343
Fjobother	-0.296792	0.577217	-0.514	0.607611
Fjobservices	-0.311595	0.593628	-0.525	0.600148
Fjobteacher	-0.321205	0.712695	-0.451	0.652628
reasonhome	-0.431435	0.319907	-1.349	0.178755
reasonother	0.159612	0.454480	0.351	0.725755
reasonreputation	-0.051845	0.317894	-0.163	0.870589
guardianmother	0.267462	0.311371	0.859	0.391226
guardianother	-0.157335	0.554872	-0.284	0.777003
traveltime	0.274301	0.197865	1.386	0.166968
studytime	-0.140650	0.155149	-0.907	0.365577
failures	-0.185333	0.211040	-0.878	0.380739
schoolsupyes	0.562716	0.379303	1.484	0.139268
famsupyes	0.369402	0.268848	1.374	0.170745
paidyes	0.060643	0.270971	0.224	0.823107
activitiesyes	-0.286006	0.247519	-1.155	0.249063
nurseryyes	-0.426858	0.315064	-1.355	0.176774
higheryes	0.503353	0.677346	0.743	0.458148
internetyes	-0.097405	0.331040	-0.294	0.768834
romanticyes	-0.243837	0.268414	-0.908	0.364577
famrel	0.494479	0.136663	3.618	0.000363 ***
freetime	-0.139869	0.138297	-1.011	0.312879
goout	0.078871	0.128794	0.612	0.540879
Dalc	-0.248633	0.178366	-1.394	0.164651
Walc	0.221434	0.139178	1.591	0.112950
health	0.027495	0.095100	0.289	0.772748
absences	0.060830	0.017915	3.396	0.000804 ***
G1	0.162966	0.076956	2.118	0.035255 *
G2	0.994677	0.066450	14.969	< 2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.868 on 235 degrees of freedom
```

```
Multiple R-squared:  0.8616, Adjusted R-squared:  0.8374
```

```
F-statistic: 35.68 on 41 and 235 DF, p-value: < 2.2e-16
```

Model Interpretation

Understanding requires a general understanding of statistics, check out Wikipedia for overviews on some of these topics, as well as the ISLR book. Here's a quick guide on understanding the model summary:

#	Name	Description
1	Residuals	The residuals are the difference between the actual values of the variable you're predicting and predicted values from your regression--$y - \hat{y}$. For most regressions you want your residuals to look like a normal distribution when plotted. If our residuals are normally distributed, this indicates the mean of the difference between our predictions and the actual values is close to 0 (good) and that when we miss, we're missing both short and long of the actual value, and the likelihood of a miss being far from the actual value gets smaller as the distance from the actual value gets larger. Think of it like a dartboard. A good model is going to hit the bullseye some of the time (but not everytime). When it doesn't hit the bullseye, it's missing in all of the other buckets evenly (i.e. not just missing in the 16 bin) and it also misses closer to the bullseye as opposed to on the outer edges of the dartboard.
2	Significance Stars	The stars are shorthand for significance levels, with the number of asterisks displayed according to the p-value computed. *** for high significance and * for low significance. In this case, *** indicates that it's unlikely that no relationship exists b/w absences and G3 scores.
3	Estimated Coefficient	The estimated coefficient is the value of slope calculated by the regression. It might seem a little confusing that the Intercept also has a value, but just think of it as a slope that is always multiplied by 1. This number will obviously vary based on the magnitude of the variable you're inputting into the regression, but it's always good to spot check this number to make sure it seems reasonable.
4	Standard Error of the Coefficient Estimate	Measure of the variability in the estimate for the coefficient. Lower means better but this number is relative to the value of the coefficient. As a rule of thumb, you'd like this value to be at least an order of magnitude less than the coefficient estimate.
5	t-value of the Coefficient Estimate	Score that measures whether or not the coefficient for this variable is meaningful for the model. You probably won't use this value itself, but know that it is used to calculate the p-value and the significance levels.
6	Variable p-value	Probability the variable is <i>NOT</i> relevant. You want this number to be as small as possible. If the number is <i>really</i> small, R will display it in scientific notation.
7	Significance Legend	The more punctuation there is next to your variables, the better. Blank=bad, Dots=pretty good, Stars=good, More Stars=very good
8	Residual Std Error / Degrees of Freedom	The Residual Std Error is just the standard deviation of your residuals. You'd like this number to be proportional to the quantiles of the residuals in #1. For a normal distribution, the 1st and 3rd quantiles should be 1.5 +/- the std error. The Degrees of Freedom is the difference between the number of observations included in your training sample and the number of variables used in your model (intercept counts as a variable).
9	R-squared	Metric for evaluating the goodness of fit of your model. Higher is better with 1 being the best. Corresponds with the amount of variability in what you're predicting that is explained by the model. WARNING: While a high R-squared indicates good correlation, correlation does not always imply causation.
10	F-statistic & resulting p-value	Performs an F-test on the model. This takes the parameters of our model (in our case we only have 1) and compares it to a model that has fewer parameters. In theory the model with more parameters should fit better. If the model with more parameters (your model) doesn't perform better than the model with fewer parameters, the F-test will have a high p-value (probability <i>NOT</i> significant boost). If the model with more parameters is better than the model with fewer parameters, you will have a lower p-value. The DF, or degrees of freedom, pertains to how many variables are in the model. In our case there is one variable so there is one degree of freedom.

Looks like Absences, G1, and G2 scores are good predictors. With age and activities also possibly contributing to a good model.

Visualize our Model

We can visualize our linear regression model by plotting out the residuals, the residuals are basically a measure of how off we are for each point in the plot versus our model (the error).

In [17]:

```
# Grab residuals
```

```
res <- residuals(model)

# Convert to DataFrame for ggplot
res <- as.data.frame(res)

head(res)
```

Out[17]:

	res
1	0.9678451
5	1.182998
6	-1.409605
7	0.1125706
9	0.381467
10	0.3221204

Why Plot Residuals?

We want a histogram of our residuals to be normally distributed, something with a strong bimodal distribution may be a warning that our data was not a good fit for linear regression. However, this can also be hidden from our model. A famous example is

[Anscombe's Quartet](#)

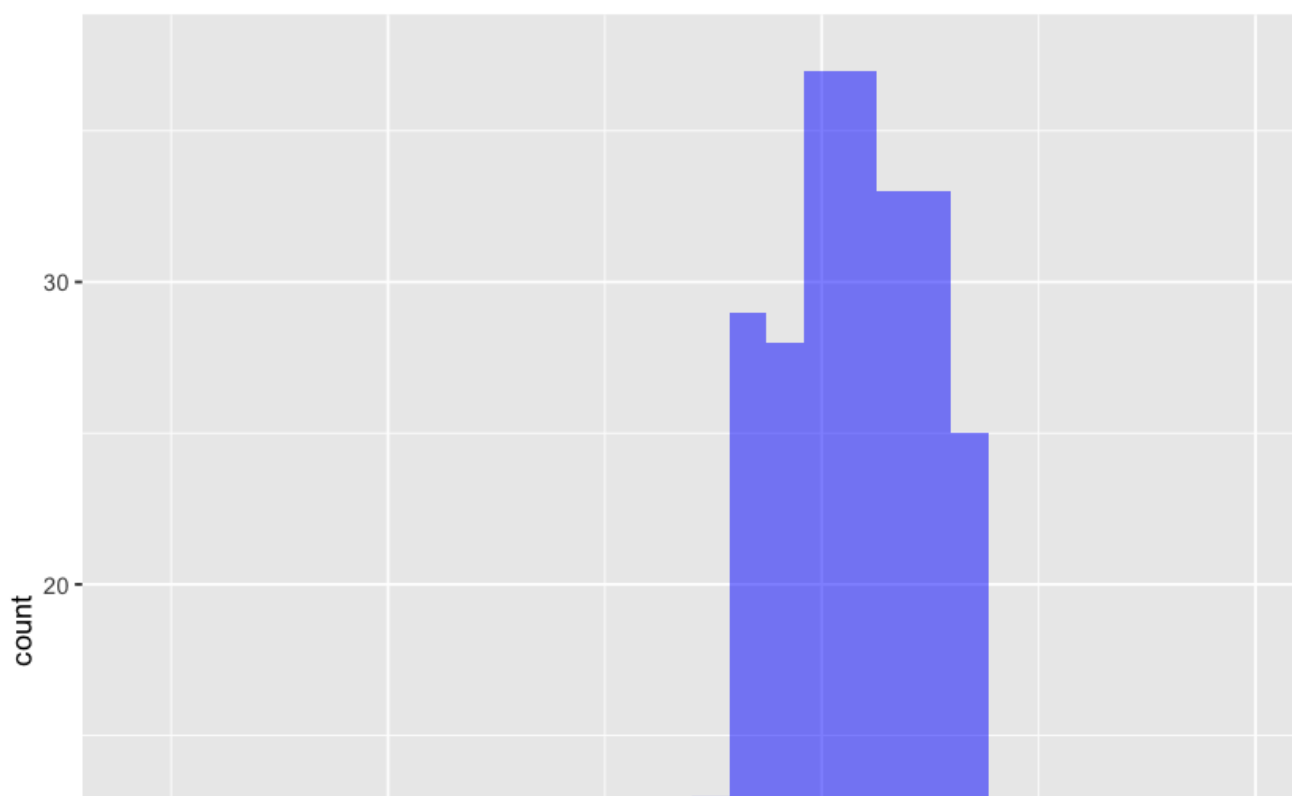
□

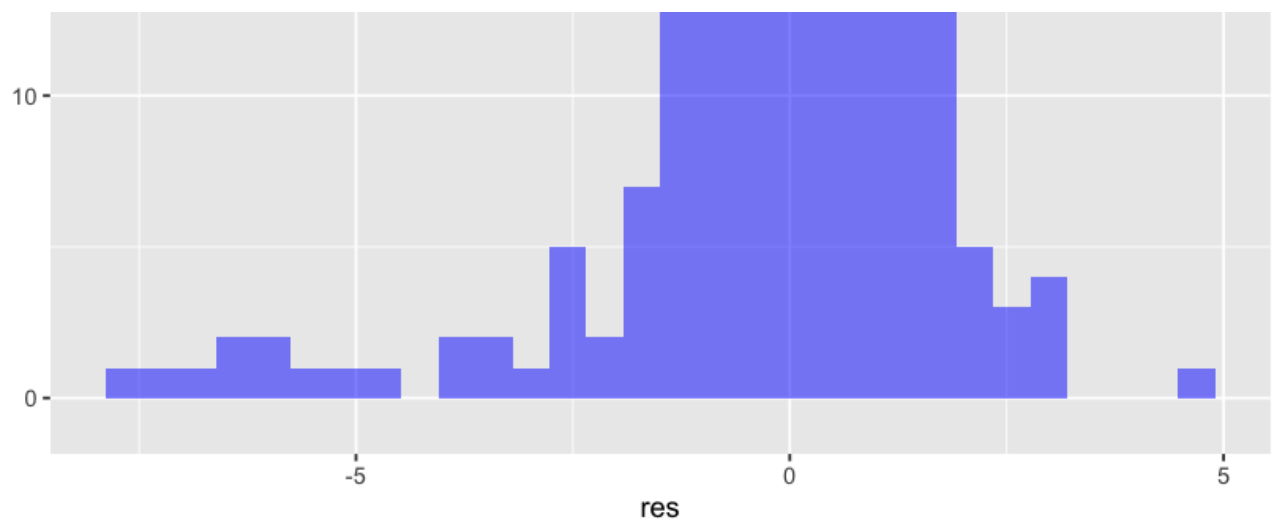
Using ggplot

In [18]:

```
# Histogram of residuals
ggplot(res, aes(res)) + geom_histogram(fill='blue', alpha=0.5)
```

`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.

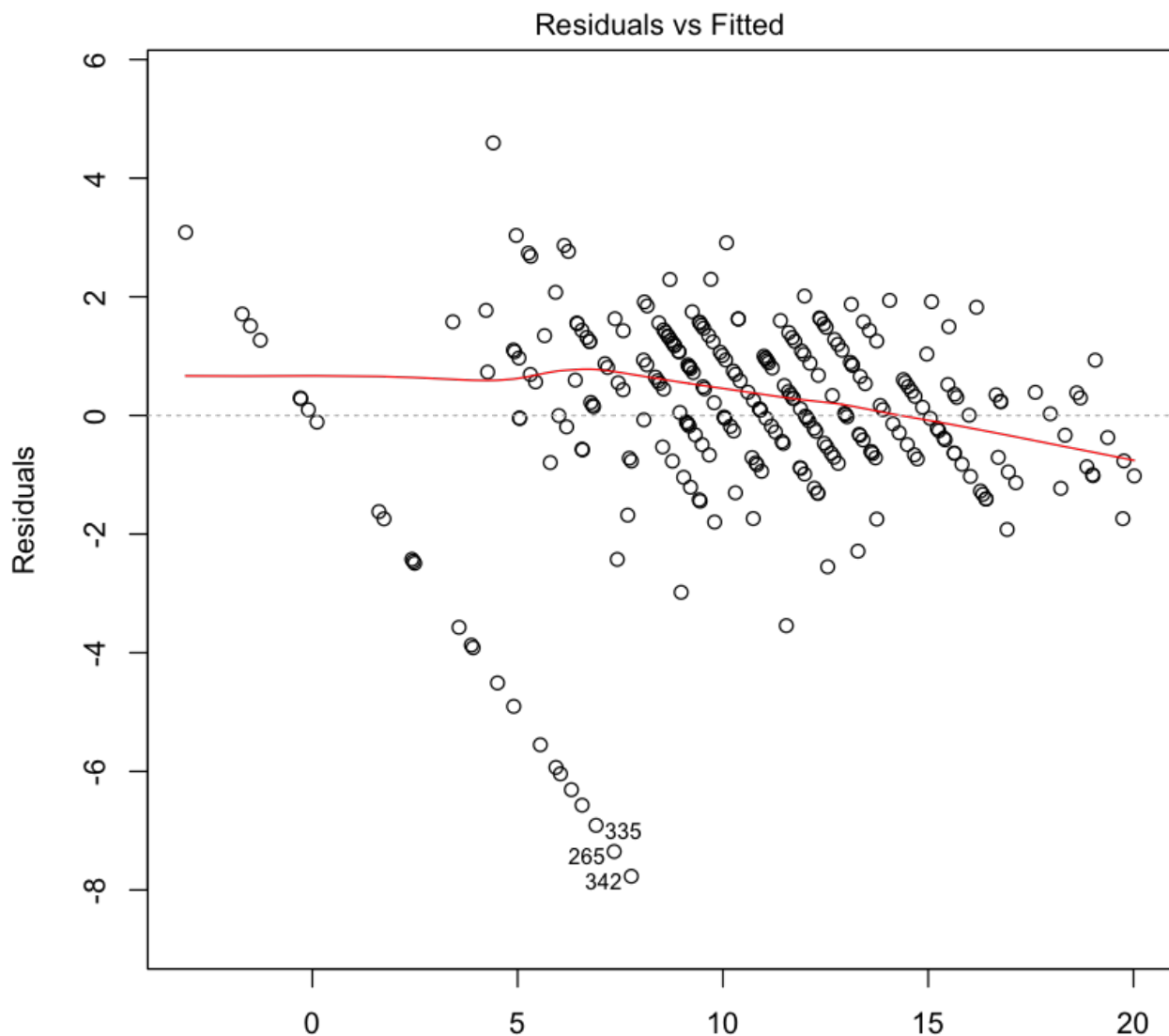




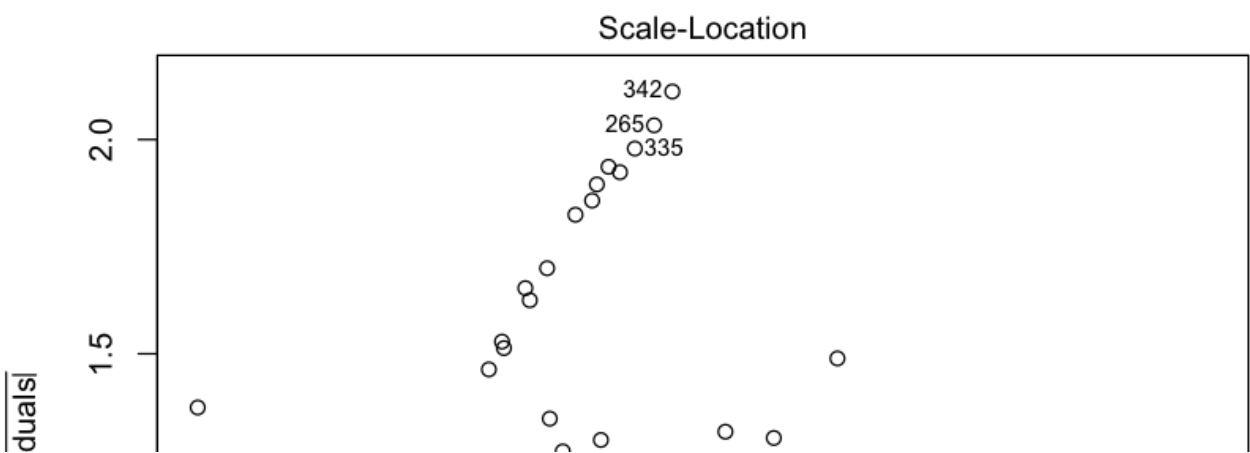
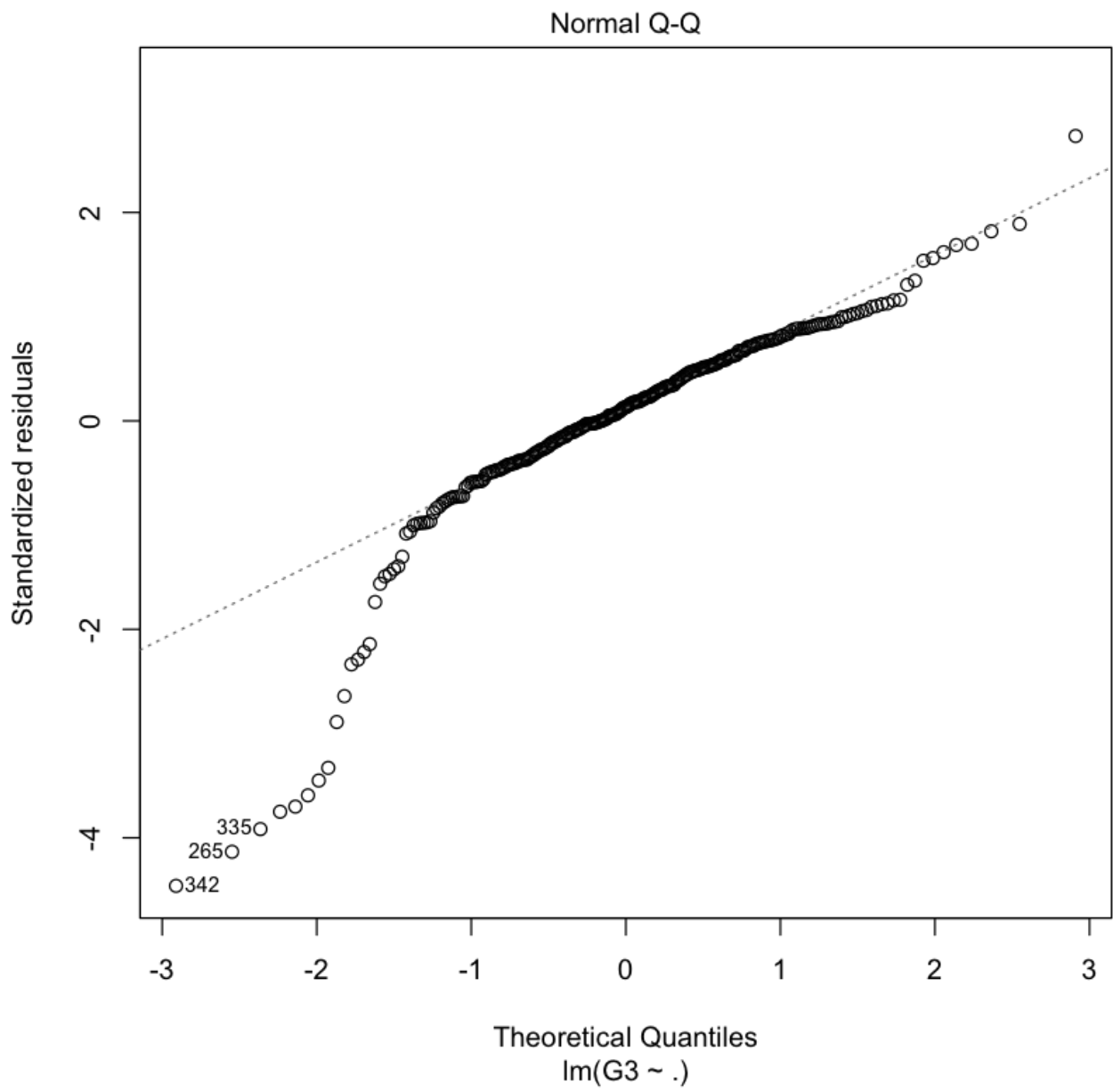
Looks like there are some suspicious residual values that have a value less than -5. We can further explore this by just calling `plot` on our model. What these plots represent is outside the course of this lecture, but it's covered in ISLR, as well as the Wikipedia page on [Regression Validation](#).

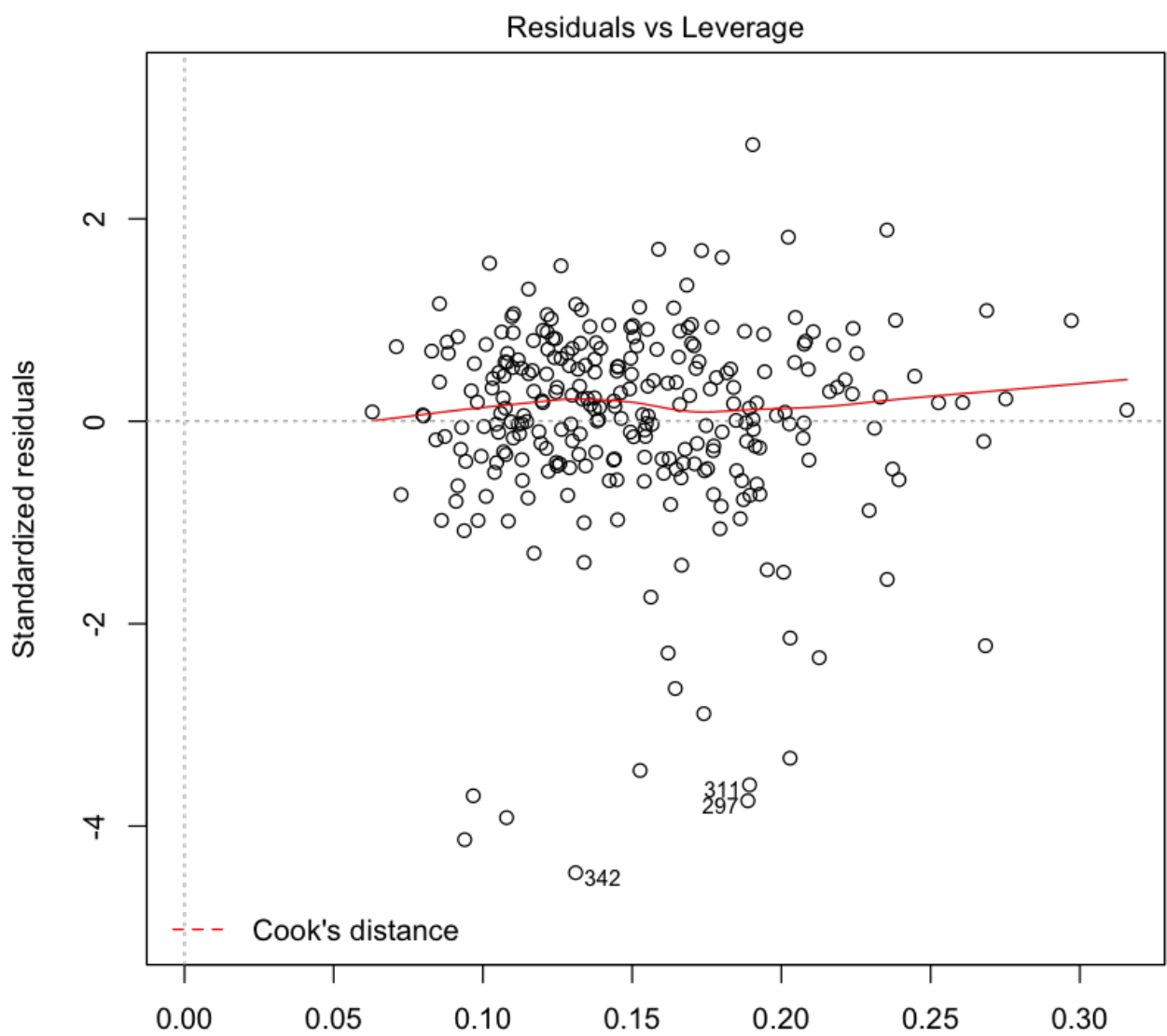
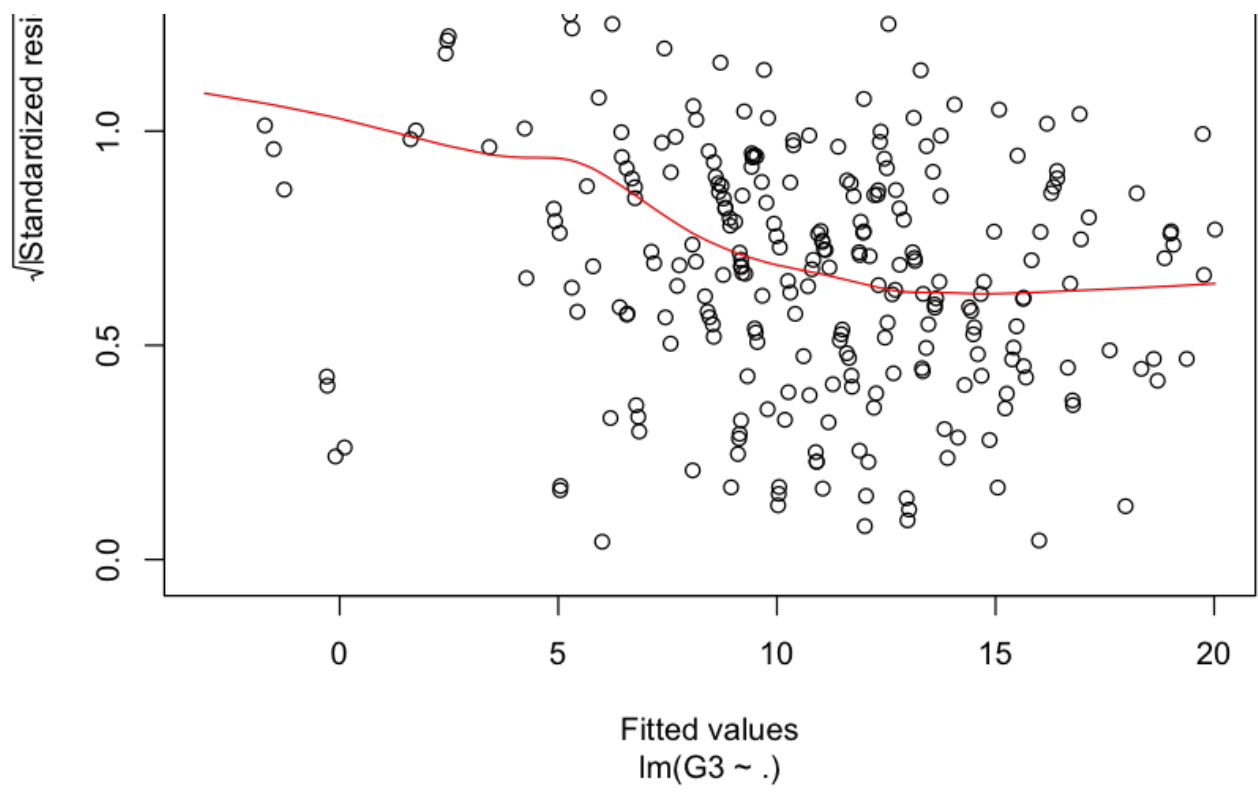
In [19]:

```
plot(model)
```



Fitted values
lm(G3 ~ .)





Leverage

lm(G3 ~ .)

Basically after looking at these plots what you will realize is that our model (behaving as a continuous line, predicted students would get **negative** scores on their test! Let's make these all zeros when running our results against our predictions.

Predictions

Let's test our model by predicting on our testing set:

In [20]:

```
G3.predictions <- predict(model, test)
```

Now we can get the root mean squared error, a standardized measure of how off we were with our predicted values:

In [21]:

```
results <- cbind(G3.predictions, test$G3)
colnames(results) <- c('pred', 'real')
results <- as.data.frame(results)
```

Now let's take care of negative predictions! Lot's of ways to this, here's a more complicated way, but its a good example of creating a custom function for a custom problem:

In [22]:

```
to_zero <- function(x) {
  if (x < 0) {
    return(0)
  } else {
    return(x)
  }
}
```

In [23]:

```
results$pred <- sapply(results$pred, to_zero)
```

There's lots of ways to evaluate the prediction values, for example the MSE (mean squared error):

In [24]:

```
mse <- mean((results$real - results$pred)^2)
print(mse)
```

```
[1] 4.411405
```

Or the root mean squared error:

In [25]:

```
mse^0.5
```

Out[25]:

```
2.10033451255583
```

Or just the R-Squared Value for our model (just for the predictions)

In [26]:

```
SSE = sum((results$pred - results$real)^2)
SST = sum( (mean(df$G3) - results$real)^2)

R2 = 1 - SSE/SST
R2
```

Out[26]:

0.777902260014962

Conclusion

You should now feel comfortable with the R syntax for a Linear Regression. If some of the plots or math did not make sense to you, make sure to review ISLR and the relevant Wikipedia pages. There is no real substitute for taking the time to read about this material

Up next is an exercise to test your knowledge!